Journal of Inequalities and Applications

RESEARCH Open Access

Higher order monotonicity involving the Mills ratio with applications

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Communicated by Árpád Baricz

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Abstract

Let $R(x)$ be the Mills ratio of the standard Gaussian law. This paper investigates the higher order monotonicity of the function

$$x \mapsto Y_{a,b}(x) = \left(x^2 + a\right) R^2(x) + bxR(x)$$

on $(0,\infty)$, where $a, b \in \mathbb{R}$. In particular, we prove that the function $x \mapsto Y_{a,a-2}(x) - a + 1$ is completely monotonic on $(0,\infty)$ if and only if $a \geq -1$. As a special case, $x \mapsto Y_{-1,-3}(x) + 2$ is completely monotonic on $(0,\infty)$, which implies the complete monotonicity of the functions $R^3(1/R)''$ and $-R^3(\ln R)'''$ on the same **interval.** As applications, several functional inequalities are derived, and some known and new sharp bounds for R are established.

Keywords: Standard Gaussian law Mills ratio; Higher order monotonicity; Inequality

Mathematics Subject Classification: Primary 44A10; 60E05; 26A48; secondary 26D15

1 Introduction

We begin with recalling the definitions of incompletely and completely monotonic functions: A function g is said to be incompletely monotonic (**IM**) on an interval I if g has derivatives of all orders on I and there exists an integer $n \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}$ such that

$$(-1)^n g^{(n)}(x) \geq 0 \text{ for } x \in I,$$

and the maximum such n is called the order of the incompletely monotonic function $g(x)$, denoted by

$$O_{\text{incm}}\left[g(I)\right] = \sup_{n \geq 0}\left\{n : (-1)^n g^{(n)}(x) \geq 0 \text{ for } x \in I\right\} \text{ (see [1, Definition 2])}$$

If $O_{\text{incm}}\left[g(I)\right] = \infty$, then $g(x)$ is said to be completely monotonic **(CM)** of I (see [2]). By the famous Bernstein Theorem [2, p. 161, Theorem 12b], a function $g(x)$ is CM on $(0,\infty)$

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if and only if

$$g(x) = \int_0^\infty e^{-xt} d\mu(t),$$

where $\mu(t)$ is nondecreasing and the integral converges for $0 < x < \infty$.

Let $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ be the density function of the standard Gaussian distribution, and $\Phi(x) = \int_{-\infty}^{x} \phi(t)\, dt$ is its cumulative distribution function. The Mills ratio [3] is defined by

$$R \equiv R(x) = \frac{1 - \Phi(x)}{\phi(x)} = e^{x^2/2} \int_x^\infty e^{-t^2/2} dt, x \geq 0,$$

which satisfies the differential equation of first order:

$$R' = xR - 1. \tag{1.1}$$

The Mills ratio finds wide applications in probability and statistics, mathematical analysis, and numerical analysis.

In 1954, Shenton [4, Eq. (20)] (see also [5]) showed that

$$R(x) = \int_0^\infty e^{-t^2/2} e^{-xt} dt, \tag{1.2}$$

implying that $R(x)$ is CM on $(0, \infty)$. Sampford [6] (see also [7], [8, Theorem 9]) proved $0 < (1/R(x))' < 1$ and $(1/R(x))'' > 0$ for $x \in \mathbb{R}$; from this, Baricz [5] pointed out that $(-1)^k [\ln R(x)]^{(k)} > 0$ for $x \in \mathbb{R}$ and $k = 1, 2, 3$, hence,

$$O_{\text{incm}} [\ln R(\mathbb{R})] \geq 3.$$

Pinelis [9] studied the monotonicity properties of the relative error of a Padé approximation for $R(x)$. Recently, Yang and Tian [10] showed that function

$$\mathcal{F}_n(x) = \frac{R_{n-1}(x) R_{n+1}(x)}{R_n(x)^2} \text{ for } n \in \mathbb{N}$$

is strictly increasing on $(0, \infty)$, where $R_n(x) = (-1)^n R^{(n)}(x)$.

In 1941, Gordon [11] first showed the double inequality

$$\frac{x}{x^2 + 1} < R(x) < \frac{1}{x} \tag{1.3}$$

for $x > 0$. Birnbaum [12] improved Gordon's lower bound a year later in (1.3) and obtained

$$\frac{2}{\sqrt{x^2 + 4} + x} < R(x) < \frac{1}{x} \tag{1.4}$$

for $x > 0$. Since then, scholars have sought bounds for $R(x)$ in the form

$$\phi_{p,q,r}(x) = \frac{p}{\sqrt{x^2 + q} + rx} \text{ with } p, q > 0, r \geq 0 \tag{1.5}$$

(see for example, [6, 13], [14, p. 179], [8, 15, 16]). It was found that the sharpest lower bounds are

$$L_B(x) = \frac{\pi}{\sqrt{x^2+2\pi}+(\pi-1)x} \text{ and } L_{GU}(x) = \frac{\pi}{\sqrt{2(4-\pi)x^2+2\pi}+2x}, \tag{1.6}$$

and the sharpest upper bounds are

$$U_S(x) = \frac{4}{\sqrt{x^2+8}+3x} \text{ and } U_B(x) = \frac{\pi}{\sqrt{(\pi-2)^2x^2+2\pi}+2x}, \tag{1.7}$$

where $L_B(x)$ and $U_B(x)$ are due to Boyd [13], $U_S(x)$ was given by Sampford [6], and $L_{GU}(x)$ was presented by Gasull and Utzet [8].

For other types of bounds and inequalities, please refer to [5, 8, 10, 17–19].

Note that all bounds of the form $\phi_{p,q,r}(x)$ can be derived from the inequality

$$Y_{a,b}(x) := \left(x^2+a\right)R^2(x) + bxR(x) < (>)\, c \tag{1.8}$$

for all $x > 0$, $a, b \in \mathbb{R}$, where

$$c = \lim_{x\to 0} Y_{a,b}(x) = \frac{a\pi}{2} \text{ or } c = \lim_{x\to\infty} Y_{a,b}(x) = b+1.$$

Therefore, the monotonicity of the function $Y_{a,b}(x)$ in x on $(0,\infty)$ will help us obtain those bounds in the form of (1.5). To ensure that such bounds have higher accuracy as $x \to \infty$ and $x \to 0$, it is required that

$$\lim_{x\to\infty} \frac{Y_{a,b}(x)-(b+1)}{x^{-2}} = a-b-2 = 0$$

and

$$\lim_{x\to 0} \frac{Y_{a,b}(x)-a\pi/2}{x} = \sqrt{\frac{\pi}{2}}(b-2a) = 0,$$

respectively. Thus, the monotonicity of $Y_{a,b}(x)$ in x on $(0,\infty)$ in the cases $b = a-2$ and $b = 2a$ will enable us to achieve the best bounds for the Mills ratio $R(x)$ on $(0,\infty)$.

Further, the higher order monotonicity of $Y_{a,b}(x)$ in x on $(0,\infty)$ in general case and two special cases $b = a-2$ and $b = 2a$ allows to establish more accurate bounds for the Mills ratio $R(x)$ for small x or large x.

The main objective of this paper is to investigate the higher order monotonicity of $Y_{a,b}(x)$ (presented in Sect. 4). To this end, we collect necessary tools in Sect. 2, establish the asymptotic expansion, Maclaurin series, and integral representation of $Y_{a,b}(x)$ in Sect. 3, and present functional inequalities and bounds for $R(x)$ as applications in Sects. 5 and 6.

2 Preliminaries and tools

To establish the asymptotic expansion and Maclaurin series of $Y_{a,b}(x)$, the following two properties of $R(x)$ are needed.

(i) Laplace's [20] (see also [21, p. 44]) asymptotic expansion

$$R(x) \sim \frac{1}{\sqrt{\pi}} \sum_{n=1}^{\infty} \frac{(-1)^{n-1} 2^{n-1} \Gamma(n-1/2)}{x^{2n-1}} \text{ as } x \to \infty. \tag{2.1}$$

A more precise asymptotic expansion was later established by Ruben [22].

(ii) Maclaurin series [8, Lemma 7]:

$$R(x) = \sum_{n=0}^{\infty} (-1)^n \frac{2^{(n-1)/2} \Gamma(n/2+1/2)}{n!} x^n. \tag{2.2}$$

To prove Property 1, we need a basic property of the hypergeometric function $F(a,b;c;x)$ defined by [23, p. 556]

$$F(a,b;c;x) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{x^n}{n!}$$

for $x \in (-1,1)$, where $a, b, c \in \mathbb{R}$ with $c \neq 0, -1, -2, \ldots$, and $(a)_n$ denotes the Pochhammer symbol defined by

$$(a)_n = a(a+1)\cdots(a+n-1) = \frac{\Gamma(n+a)}{\Gamma(a)},$$

for $n = 1, 2, \ldots$, and $(a)_0 = 1$ for $a \neq 0$. Here $\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$ $(x > 0)$ is the gamma function. In the case of $a + b < c$, we have

$$F(a,b;c;1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)} \text{ ([23, Eq. (15.1.20)])} \tag{2.3}$$

To prove our main results, we need the following three key tools. The first tool is the convolution formula for the Laplace transforms.

Lemma 2.1 ([24, Theorem 2.39]) *Let $f_i(t)$ for $i = 1, 2$ be piecewise continuous in arbitrary finite intervals included on $(0, \infty)$. If there exist some constants $M_i > 0$ and $c_i \geq 0$ such that $|f_i(t)| \leq M_i e^{c_i t}$ for $i = 1, 2$, then*

$$\int_0^{\infty} f_1(u) e^{-su} du \int_0^{\infty} f_2(v) e^{-sv} dv = \int_0^{\infty} \left(\int_0^t f_1(u) f_2(t-u) du \right) e^{-st} dt. \tag{2.4}$$

The second tool is the sign rule for the NP(PN)-type power series. A real sequence is called positive (negative) if every term is nonnegative (nonpositive) with at least one nonzero term. A real sequence $\{a_n\}_{n\geq 0}$ is negative-positive (NP)-type, if there exists an $m \in \mathbb{N}$ such that $\{a_n\}_{0\leq n\leq m}$ is negative and $\{a_n\}_{n>m}$ is positive. The sequence $\{-a_n\}_{n\geq 0}$ is called positive-negative (PN)-type. In 2021, Yang, Tian and Zhu [25, Definition 1] first introduced the definition of NP and PN-type power series.

Definition 2.1 An NP-type power series is defined as $S(t) = \sum_{k=0}^{\infty} a_k t^k$, **where** the sequence $\{a_n\}_{n\geq 0}$ is NP-type; $-S(t)$ is a PN-type power series.

In 2017, Yang and Tian [26, Lemma 2] established the following signs rule (see also [27, 28]).

Lemma 2.2 *Let $S(t)$ be an NP-type power series converging on the interval $(0,r)$ $(r>0)$.*

(i) If $S(r^-) \le 0$, then $S(t) < 0$ for all $t \in (0,r)$.

(ii) If $S(r^-) > 0$, then there is a unique $t_0 \in (0,r)$ such that $S(t) < 0$ for $t \in (0,t_0)$ and $S(t) > 0$ for $t \in (t_0,r)$.

In particular, if the NP-type power series $S(t)$ converges on $(0,\infty)$, then there is a unique $t_0 \in (0,\infty)$ such that $S(t) < 0$ for $t \in (0,t_0)$ and $S(t) > 0$ for $t \in (t_0,\infty)$.

The third tool is the sign rule for the Laplace integrals, which appeared in [1].

Lemma 2.3 *Let $t_0 > 0$, and let the functions f be integrable on $(0,\infty)$ with $f(t) > (<)\, 0$ for $t \in [0,t_0)$ and $f(t) < (>)\, 0$ for $t \in (t_0,\infty)$. Suppose that $F(x) = \int_0^\infty f(t)\, e^{-xt} dt$ converge for $0 < x < \infty$. The following statements are valid.*

(i) The inequality

$$F(x) = \int_0^\infty f(t)\, e^{-xt} dt \ge (\le)\, 0$$

holds for $x \in (0,\infty)$ if and only if $F(0^+) = \int_0^\infty f(t)\, dt \ge (\le)\, 0$.

(ii) If $F(0^+) = \int_0^\infty f(t)\, dt < (>)\, 0$, then there exists a unique $x_0 \in (0,\infty)$ such that $F(x) < (>)\, 0$ for $x \in (0,x_0)$ and $F(x) > (<)\, 0$ for $x \in (x_0,\infty)$.

Remark 2.1 From Lemma 2.3, if there exists an $x_1 > 0$ such that $F(x_1) < (>)\, 0$, then $F(x) < (>)\, 0$ for all $x \in (0,x_1)$; if $F(x_1) > (<)\, 0$, then $F(x) > (<)\, 0$ for all $x \in (x_1,\infty)$.

3 Asymptotic expansion, Maclaurin series and integral representation

This section establishes the asymptotic expansion, Maclaurin series, and integral representation of $Y_{a,b}(x)$.

3.1 Asymptotic expansion

Lemma 3.1 *As $x \to \infty$, $R^2(x)$ has the asymptotic expansion*:

$$R^2(x) \sim \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{\alpha_n}{x^{2n}}, \tag{3.1}$$

where

$$\alpha_n = (-1)^{n-1} \sqrt{\pi}\, (n-1)! \sum_{k=0}^{n-1} \frac{2^k \Gamma(k+1/2)}{k!}. \tag{3.2}$$

Proof Differentiating both sides of (3.1) gives

$$\left[R^2(x)\right]' = 2R(x) R'(x) \sim \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{-2n\alpha_n}{x^{2n+1}} = \sum_{n=2}^{\infty} \frac{-2(n-1)\alpha_{n-1}}{\pi} \frac{1}{x^{2n-1}} \text{ as } x \to \infty. \tag{3.3}$$

Using $R'(x) = xR(x) - 1$ (from (1.1), we have

$$\left[R^2(x)\right]' = 2R(x)R'(x) = 2xR^2(x) - 2R(x). \tag{3.4}$$

Substituting the asymptotic expansions of (3.1) and (2.1) yields, as $x \to \infty$,

$$\begin{aligned} 2R(x)R'(x) = 2xR^2(x) - 2R(x) &\sim \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{x\alpha_n}{x^{2n}} - 2\frac{1}{\sqrt{\pi}}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}2^{n-1}\Gamma(n-1/2)}{x^{2n-1}} \\ &= \sum_{n=1}^{\infty}\frac{2\alpha_n - (-1)^{n-1}\sqrt{\pi}2^n\Gamma(n-1/2)}{\pi}\frac{1}{x^{2n-1}}. \end{aligned} \tag{3.5}$$

Equating coefficients in (3.3) and (3.5) gives $2\alpha_1 - \sqrt{\pi}2\Gamma(1/2) = 0$ and for $n \geq 2$,

$$-2(n-1)\alpha_{n-1} = 2\alpha_n - (-1)^{n-1}\sqrt{\pi}2^n\Gamma(n-1/2),$$

which can be written as

$$(-1)^n\frac{\alpha_n}{(n-1)!} - (-1)^{n-1}\frac{\alpha_{n-1}}{(n-2)!} = -\sqrt{\pi}\frac{2^{n-1}\Gamma(n-1/2)}{(n-1)!}.$$

Solving the recurrence equation gives (3.2), completing the proof. □

Theorem 3.1 *Let $Y_{a,b}(x)$ be defined in* **(1.8)**. *Then*

$$Y_{a,b}(x) \sim b + 1 + \sum_{n=1}^{\infty}\frac{(-1)^n u_n}{x^{2n}} \text{ as } x \to \infty, \tag{3.6}$$

where

$$u_n \equiv u_n(a,b) = (b+1)\frac{2^n\Gamma(n+1/2)}{\Gamma(1/2)} + (n-a)(n-1)!\sum_{k=0}^{n-1}\frac{2^k\Gamma(k+1/2)}{\Gamma(1/2)k!}. \tag{3.7}$$

In particular, $u_1 = b + 2 - a$, $u_2 = 3b + 7 - 2a$, $u_3 = 15b + 36 - 7a$, and $u_4 = 105b - 36a + 249$.

Remark 3.1 (i) Putting $u_1 = b + 2 - a = 0$ yields

$$Y_{a,a-2}(x) \sim a - 1 + \frac{a+1}{x^4} - \frac{2(4a+3)}{x^6} + \frac{3(23a+13)}{x^8} + \cdots \text{ as } x \to \infty. \tag{3.8}$$

(ii) Letting $u_1 = b + 2 - a = 0$ and $u_2 = 3b + 7 - 2a = 0$ gives $(a,b) = (-1,-3)$. Then

$$Y_{-1,-3}(x) \sim -2 + \frac{2}{x^6} - \frac{30}{x^8} + \frac{396}{x^{10}} - \frac{5460}{x^{12}} + \cdots \text{ as } x \to \infty. \tag{3.9}$$

Proof of Theorem 3.1 Substituting the asymptotic expansions (3.1) (Lemma 3.1) and (2.1) into $Y_{a,b}(x) = x^2R^2(x) + aR^2(x) + bxR(x)$, then Simplifying and collecting like terms give

$$Y_{a,b}(x) \sim \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\alpha_n}{x^{2n-2}} + a\frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\alpha_n}{x^{2n}} + bx\frac{1}{\sqrt{\pi}}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}2^{n-1}\Gamma(n-1/2)}{x^{2n-2}}$$

$$= b + 1 + \sum_{n=1}^{\infty} \frac{\alpha_{n+1} + a \times \alpha_n + b\sqrt{\pi}\,(-1)^n\, 2^n \Gamma\,(n + 1/2)}{\pi x^{2n}}$$

$$= b + 1 + \sum_{n=1}^{\infty} \frac{(-1)^n\, u_n}{x^{2n}} \text{ as } x \to \infty,$$

completing the proof. □

3.2 Maclaurin series

Lemma 3.2 *Let*

$$R^2\,(x) = \sum_{n=0}^{\infty} \beta_n x^n. \tag{3.10}$$

Then $\beta_0 = \pi/2$, $\beta_1 = -\sqrt{2\pi}$, $\beta_2 = \pi/2 + 1$ *and for* $n \geq 1$,

$$\beta_n = \begin{cases} -\frac{\sqrt{2\pi}}{\Gamma(m+1/2)} \sum_{k=0}^{m-1} \frac{\Gamma(k+1/2)}{2^{k+1}k!} & \text{if } n = 2m-1, \\ \\ \frac{\pi}{2}\frac{1}{m!} + \frac{\sqrt{\pi}}{m!} \sum_{k=1}^{m} \frac{(k-1)!}{2^k \Gamma(k+1/2)} & \text{if } n = 2m. \end{cases} \tag{3.11}$$

Proof Using the Cauchy product formula on the Maclaurin series of $R\,(x)$ (2.2), we derive

$$\beta_n = (-1)^n\, 2^{n/2-1} \sum_{k=0}^{n} \frac{\Gamma\,((k+1)/2)\,\Gamma\,((n-k+1)/2)}{k!\,(n-k)!}.$$

It follows that $\beta_0 = \pi/2$, $\beta_1 = -\sqrt{2\pi}$ and $\beta_2 = \pi/2 + 1$. By (3.4) and (3.10), we have

$$\left[R^2\,(x)\right]' = 2R\,(x)\,R'\,(x) = 2xR^2\,(x) - 2R\,(x) = \sum_{n=0}^{\infty} n\beta_n x^{n-1}.$$

Then

$$2\sum_{n=1}^{\infty} \beta_{n-1} x^n - 2\sum_{n=0}^{\infty} \frac{(-1)^n\, 2^{(n-1)/2} \Gamma\,((n+1)/2)}{n!} x^n = \sum_{n=0}^{\infty} (n+1)\,\beta_{n+1} x^n.$$

Comparing the coefficient of x^n gives the recurrence relation

$$(n+1)\,\beta_{n+1} - 2\beta_{n-1} = -2\frac{(-1)^n\, 2^{(n-1)/2} \Gamma\,((n+1)/2)}{n!}. \tag{3.12}$$

We solve this recurrence equation by considering two cases.

Case 1: n is an even number. Then there exists a positive integer m such that $n = 2m$, $m \in \mathbb{N}$. By the duplication formula [23, Eq. (6.1.18)]

$$\Gamma\,(2x) = \frac{2^{2x-1}}{\sqrt{\pi}} \Gamma\left(x + \frac{1}{2}\right) \Gamma\,(x), \tag{3.13}$$

(3.12) becomes

$$(2m+1)\,\beta_{2m+1} - 2\beta_{2m-1} = -\frac{2^{m+1/2} \Gamma\,(m+1/2)}{(2m)!} = -\frac{\sqrt{2\pi}}{2^m m!},$$

which can be written as

$$\Gamma\left(m+\frac{3}{2}\right)\beta_{2m+1}-\Gamma\left(m+\frac{1}{2}\right)\beta_{2m-1}=-\frac{\sqrt{2\pi}\,\Gamma\left(m+1/2\right)}{2^{m+1}m!}.$$

Solving with $\beta_1=-\sqrt{2\pi}$ gives (3.11) for $n=2m-1$.

Case 2: n is an odd number. Then there exists a positive integer m such that $n=2m-1$, $m\in\mathbb{N}$. By the duplication formula (3.13), (3.12) becomes

$$m!\beta_{2m}-(m-1)!\beta_{2m-2}=\frac{2^{m-1}\Gamma\left(m\right)^2}{\Gamma\left(2m\right)}=\sqrt{\pi}\,\frac{\Gamma\left(m\right)}{2^m\Gamma\left(m+1/2\right)}.$$

Solving with $\beta_0=\pi/2$ yields (3.11) for $n=2m$.

This completes the proof. □

Theorem 3.2 *For $Y_{a,b}(x)$ defined in (1.8), the Maclaurin series is*:

$$Y_{a,b}(x)=\left(x^2+a\right)R^2(x)+bxR(x)=\sum_{n=0}^{\infty}v_nx^n, \tag{3.14}$$

where $v_0=a\pi/2$ and $v_n\equiv v_n(a,b)$ for $n\geq 1$ is given by

$$v_n=\begin{cases}\frac{\sqrt{2\pi}(b+1)}{2^m(m-1)!}-\frac{\sqrt{2\pi}(m+a-1/2)}{\Gamma(m+1/2)}\sum_{k=0}^{m-1}\frac{\Gamma(k+1/2)}{2^{k+1}k!} & \text{if } n=2m-1,\\ \frac{\pi}{2}\frac{m+a}{m!}-\frac{\sqrt{\pi}(b+1)}{2^m\Gamma(m+1/2)}+\frac{(m+a)\sqrt{\pi}}{m!}\sum_{k=1}^{m}\frac{(k-1)!}{2^k\Gamma(k+1/2)} & \text{if } n=2m.\end{cases} \tag{3.15}$$

In particular, $v_1=\sqrt{\pi/2}\,(b-2a)$, $v_2=(\pi/2+1)\,a-b+\pi/2$ and $v_3=\sqrt{2\pi}\,(3b-10a-12)/12$.

Remark 3.2 Setting $v_1=\sqrt{\pi/2}\,(b-2a)=0$ leads to

$$Y_{a,2a}(x)=\frac{a\pi}{2}+\frac{(\pi-2)\,a+\pi}{2}x^2-\frac{\sqrt{2\pi}\,(a+3)}{3}x^3+\cdots.$$

Proof of Theorem 3.2 Substituting the Maclaurin series of (3.10) and (2.2) into $Y_{a,b}(x)$ gives

$$\begin{aligned}Y_{a,b}(x)&=\sum_{n=0}^{\infty}\beta_nx^{n+2}+a\sum_{n=0}^{\infty}\beta_nx^n+b\sum_{n=0}^{\infty}(-1)^n\frac{2^{(n-1)/2}\Gamma\left(n/2+1/2\right)}{n!}x^{n+1}\\&=a\beta_0+v_1x+v_2x^2+\sum_{n=3}^{\infty}v_nx^n,\end{aligned}$$

where $v_0=a\beta_0$, $v_1=-\sqrt{\pi/2}\,(2a-b)$, $v_2=\pi/2+a\,(\pi/2+1)-b$ and for $n\geq 3$,

$$v_n=\beta_{n-2}+a\beta_n+\frac{b}{2}(-1)^{n-1}\frac{2^{n/2}\Gamma\left(n/2\right)}{(n-1)!} \tag{3.16}$$

Substituting (3.11) into (3.16) with the duplication formula (3.13) yields (3.15). This completes the proof. □

For later use, we need an important property of $v_n(a,b)$.

Property 1 Let $v_n \equiv v_n(a,b)$ be defined by (3.15). For fixed $a,b \in \mathbb{R}$, if

$$n > \max\left\{-2a, 2b-2a+2, \frac{8}{5}b-2a+\frac{8}{5}\right\} = \varphi(a,b), \tag{3.17}$$

then $(-1)^n v_n(a,b) > 0$.

Remark 3.3 By Property 1, there must exist

$$N = \lfloor \varphi(a,b) \rfloor \text{ if } \varphi(a,b) \geq 0 \text{ and } N = -1 \text{ if } \varphi(a,b) < 0$$

such that $(-1)^n v_n(a,b) > 0$ for $n > N \in \mathbb{N}_0 \cup \{-1\}$ and fixed $a,b \in \mathbb{R}$, where $\lfloor x \rfloor$ denote the largest integer that is less than or equal to x.

Proof of Property 1 We divide the proof into two cases.

Case 1: $n = 2m-1, m \in \mathbb{N}$. Then the condition (3.17) becomes

$$m > \max\left\{\frac{1}{2}-a, b-a+\frac{3}{2}, \frac{4}{5}b-a+\frac{13}{10}\right\}. \tag{3.18}$$

By (3.15), we have

$$(-1)^n v_n = -v_{2m-1} = \frac{\sqrt{2\pi}(m+a-1/2)}{\Gamma(m+1/2)} \sum_{k=0}^{m-1} \frac{\Gamma(k+1/2)}{2^{k+1}k!} - \frac{\sqrt{2\pi}(b+1)}{2^m(m-1)!},$$

which can be written as

$$-\frac{2^m(m-1)!}{\sqrt{2\pi}} v_{2m-1} = \frac{m+a-1/2}{m} \sum_{k=0}^{m-1} 2^{m-k-1} \frac{\Gamma(m+1)\Gamma(k+1/2)}{\Gamma(k+1)\Gamma(m+1/2)} - (b+1).$$

Note that for $m \geq k+1$, we have $2^{m-k-1} \geq 1$ and by (2.3),

$$\begin{aligned}\frac{\Gamma(m+1)\Gamma(k+1/2)}{\Gamma(k+1)\Gamma(m+1/2)} &= F\left(\frac{1}{2}, m-k; m+1; 1\right) = \sum_{j=0}^{\infty} \frac{(1/2)_j (m-k)_j}{(m+1)_j j!} \\ &> 1 + \frac{(1/2)(m-k)}{m+1}.\end{aligned}$$

Thus, if m satisfies the inequality (3.18), then

$$\begin{aligned}-\frac{2^m(m-1)!}{\sqrt{2\pi}} v_{2m-1} &> \frac{m+a-1/2}{m} \sum_{k=0}^{m-1} \left(1 + \frac{(1/2)(m-k)}{m+1}\right) - (b+1) \\ &= \frac{5}{4}\left(m+a-\frac{4}{5}b-\frac{13}{10}\right) > 0.\end{aligned}$$

Case 2: $n = 2m, m \in \mathbb{N}$. Then the condition (3.17) becomes

$$m > \max\left\{-a, b-a+1, \frac{4}{5}b-a+\frac{4}{5}\right\}. \tag{3.19}$$

By (3.15), we have

$$(-1)^n \nu_n = \nu_{2m} = \frac{\pi}{2}\frac{m+a}{m!} - \frac{\sqrt{\pi}\,(b+1)}{2^m\Gamma\,(m+1/2)} + \frac{(m+a)\,\sqrt{\pi}}{m!}\sum_{k=1}^{m}\frac{(k-1)!}{2^k\Gamma\,(k+1/2)},$$

which can be written as

$$\frac{2^m\Gamma\,(m+1/2)}{\sqrt{\pi}}\nu_{2m} = \sqrt{\pi}\frac{(m+a)\,2^{m-1}\Gamma\,(m+1/2)}{m!} - (b+1) + \frac{m+a}{m}\sum_{k=1}^{m}2^{m-k}\frac{\Gamma\,(m+1/2)\,\Gamma\,(k)}{\Gamma\,(k+1/2)\,\Gamma\,(m)}.$$

Note that for $m \geq k$, we have $2^{m-k} \geq 1$ and by (2.3),

$$\frac{\Gamma\,(m+1/2)\,\Gamma\,(k)}{\Gamma\,(k+1/2)\,\Gamma\,(m)} = F\left(\frac{1}{2}, m-k; m+\frac{1}{2}; 1\right) = \sum_{j=0}^{\infty}\frac{(1/2)_j\,(m-k)_j}{(m+1/2)_j\,j!} > 1.$$

Thus, if m satisfies the inequality (3.19), then

$$\frac{2^m\Gamma\,(m+1/2)}{\sqrt{\pi}}\nu_{2m} > 0 - (b+1) + \frac{m+a}{m}\sum_{k=1}^{m}1 = m+a-b-1 > 0.$$

This completes the proof. □

3.3 Integral representation

Theorem 3.3 *The function $Y_{a,b}(x)$ defined by (1.8) has the integral representation*

$$Y_{a,b}(x) = b+1+\int_0^{\infty}\eta_{a,b}(s)\,se^{-s^2/2}e^{-xs}ds, \tag{3.20}$$

where

$$\eta_{a,b}(s) = a-(b+2) + \frac{\sqrt{\pi}}{2}\sum_{n=1}^{\infty}\frac{n+a}{4^n\Gamma\,(n+3/2)}s^{2n}. \tag{3.21}$$

Proof Applying Lemma 2.1 (convolution formula) with $\xi(u) = e^{-u^2/2}$, we have

$$R^2(x) = \left(\int_0^{\infty}e^{-s^2/2}e^{-xs}ds\right)^2 = \int_0^{\infty}\left[\int_0^{s}\xi(u)\,\xi(s-u)\,du\right]e^{-xs}ds.$$

By (1.1) and (1.2),

$$xR(x) = R'(x)+1 = 1-\int_0^{\infty}se^{-s^2/2}e^{-xs}ds.$$

Thus,

$$x^2R^2(x) = 1-2\int_0^{\infty}se^{-s^2/2}e^{-xs}ds + \left(\int_0^{\infty}se^{-s^2/2}e^{-xs}ds\right)^2$$

$$= 1 - 2\int_0^\infty s\xi(s)\, e^{-xs} ds + \int_0^\infty \left[\int_0^s u(s-u)\,\xi(u)\,\xi(s-u)\, du\right] e^{-xs} ds.$$

Substituting into $Y_{a,b}(x) = x^2 R^2(x) + aR^2(x) + bxR(x)$ and simplifying give

$$Y_{a,b}(x) = b + 1 + \int_0^\infty \eta^*_{a,b}(s)\, e^{-xs} ds,$$

where

$$\eta^*_{a,b}(s) = \int_0^s [u(s-u) + a]\,\xi(u)\,\xi(s-u)\, du - (b+2)\, s\xi(s).$$

Letting $u = s(1+v)/2$, we obtain

$$u(s-u) = \frac{1}{4}s^2\left(1-v^2\right),$$
$$\xi(u)\,\xi(s-u) = \exp\left(\frac{1}{2}\left(\frac{1+v}{2}s\right)^2\right)\exp\left(\frac{1}{2}\left(\frac{1-v}{2}s\right)^2\right) = \exp\left(-\frac{\left(v^2+1\right)s^2}{4}\right),$$

and then,

$$\eta^*_{a,b}(s) = \int_{-1}^1 \left(\frac{1}{4}s^2\left(1-v^2\right) + a\right) e^{-s^2\left(v^2+1\right)/4}\frac{s}{2} dv - (b+2)\, se^{-s^2/2}$$
$$= s\int_0^1 \left(\frac{1}{4}s^2\left(1-v^2\right) + a\right) e^{-s^2\left(1+v^2\right)/4} dv - (b+2)\, se^{-s^2/2}.$$

Expanding in power series yields

$$\left(\frac{1}{4}s^2\left(1-v^2\right) + a\right) e^{s^2\left(1-v^2\right)/4} = a + \sum_{n=1}^\infty \frac{n+a}{n!}\frac{\left(1-v^2\right)^n}{4^n} s^{2n}.$$

Thus

$$\eta_{a,b}(s) = \frac{\eta^*_{a,b}(s)}{se^{-s^2/2}} = a - (b+2) + \sum_{n=1}^\infty \frac{n+a}{4^n n!} s^{2n}\int_0^1 \left(1-v^2\right)^n dv$$
$$= a - (b+2) + \frac{\sqrt{\pi}}{2}\sum_{n=1}^\infty \frac{n+a}{4^n \Gamma(n+3/2)} s^{2n},$$

where we have used the integral formula

$$\int_0^1 \left(1-v^2\right)^n dv = \frac{1}{2}B\left(\frac{1}{2}, n+1\right) = \frac{\sqrt{\pi}}{2}\frac{\Gamma(n+1)}{\Gamma(n+3/2)}.$$

This completes the proof. □

4 Higher order monotonicity

In this section, we investigate the higher order monotonicity of the function $Y_{a,b}(x)$ in x on $(0,\infty)$ in general case and two special cases: $b = a-2$ and $b = 2a$. As shown in Introduction,

the two special cases have unique significance. Precisely, the higher order monotonicity of the function $Y_{a,b}(x)$ in x on $(0,\infty)$ in the two special cases will yield the best bounds for $R(x)$.

4.1 General case

Theorem 4.1 *Let*

$$\ddot{Y}_{a,b}(x) = Y_{a,b}(x) - (b+1), \tag{4.1}$$

where $Y_{a,b}(x)$ is defined in (1.8), and let $\nu_n = \nu_n(a,b)$ be defined by (3.15). Then the following statements are valid.

(i) If $(a,b) \in \{a \geq \max\{b+2,-1\}\}$, then $x \mapsto \ddot{Y}_{a,b}(x)$ is CM on $(0,\infty)$.

(ii) When $(a,b) \in \{a < b+2\} \cup \{a = b+2 < -1\}$, $x \mapsto +(-)\ddot{Y}_{a,b}(x)$ is not CM on $(0,\infty)$. Precisely, $(-1)^n \ddot{Y}^{(n)}_{a,b}(x)(x) \leq 0$ for $x \in (0,\infty)$ and $n \in \mathbb{N}_0$ if and only if

$$\nu_0(a,b) - (b+1) \leq 0 \text{ if } n = 0 \text{ and } (-1)^n \nu_n(a,b) \leq 0 \text{ if } n \geq 1; \tag{4.2}$$

while

$$\nu_0(a,b) - (b+1) > 0 \text{ if } n = 0 \text{ and } (-1)^n \nu_n(a,b) > 0 \text{ if } n \geq 1, \tag{4.3}$$

there is an $x_n \in (0,\infty)$ such that $(-1)^n \ddot{Y}^{(n)}_{a,b}(x) > 0$ for $x \in (0,x_n)$ and $(-1)^n \ddot{Y}^{(n)}_{a,b}(x) < 0$ for $x \in (x_n,\infty)$.

Proof By Theorem 3.3, for $n \in \mathbb{N}$,

$$(-1)^n \ddot{Y}^{(n)}_{a,b}(x) = \int_0^\infty \eta_{a,b}(s)\, s^{n+1} e^{-s^2/2} e^{-xs} ds, \tag{4.4}$$

where $\eta_{a,b}(s)$ is defined by (3.21).

(i) For $a \geq \max\{b+2,-1\}$, $\eta_{a,b}(s) > 0$ for $s > 0$. Hence, $(-1)^n \ddot{Y}^{(n)}_{a,b}(x) > 0$ for $x > 0$ and every $n \in \mathbb{N}_0$. That is, $x \mapsto \ddot{Y}_{a,b}(x)$ is CM on $(0,\infty)$.

(ii) For $(a,b) \in \{a < b+2\} \cup \{a = b+2 < -1\}$, we distinguish three cases to show that $\eta_{a,b}(s)$ is an NP-type power series.

Case 1: $-1 \leq a < b+2$. Clearly, $\eta_{a,b}(s)$ is an NP-type power series.

Case 2: $a < b+2$ and $a < -1$. Then there exists an $n_0 \geq 2$ such that $n + a < 0$ for $1 \leq n < n_0$ and $n + a > 0$ for $n > n_0$. This implies that $\eta_{a,b}(s)$ is also an NP-type power series.

Case 3: $a = b+2 < -1$. Similarly, $\eta_{a,b}(s)$ is an NP-type power series.

By Lemma 2.2 (sign rule for the NP-type power series), there is an $s_0 > 0$ such that $\eta_{a,b}(s) < 0$ for $s \in (0,s_0)$ and $\eta_{a,b}(s) > 0$ for $s \in (s_0,\infty)$. By Bernstein Theorem, the function $x \mapsto +(-)\ddot{Y}_{a,b}(x)$ is not CM on $(0,\infty)$.

Applying Lemma 2.3 (sign rule for Laplace integrals) to (4.4), we find that $(-1)^n \ddot{Y}^{(n)}_{a,b}(x) \leq 0$ for $x \in (0,\infty)$ and $n \in \mathbb{N}_0$ if and only if $(-1)^n \ddot{Y}^{(n)}_{a,b}(0) \leq 0$. Since $\ddot{Y}^{(0)}_{a,b}(0) = \nu_0 - (b+1)$ and $\ddot{Y}^{(n)}_{a,b}(0) = n!\nu_n$ if $n \geq 1$, this is equivalent to (4.2). If ν_n satisfies (4.3) for $n \in \mathbb{N}$, then there is an $x_n \in (0,\infty)$ such that $(-1)^n \ddot{Y}^{(n)}_{a,b}(x) > 0$ for $x \in (0,x_n)$ and $(-1)^n \ddot{Y}^{(n)}_{a,b}(x) < 0$ for $x \in (x_n,\infty)$.

This completes the proof. □

Remark 4.1 Remark 3.3 tells us that there must exist n_0 defined by

$$n_0 = \min\left\{N : (-1)^n \nu_n(a,b) > 0 \text{ for } n \geq N \in \mathbb{N}_0 \text{ and fixed } a, b \in \mathbb{R}\right\}. \tag{4.5}$$

Then, $(-1)^{n_0} \nu_{n_0}(a,b) > 0$ and $(-1)^{n_0-1} \nu_{n_0-1}(a,b) \leq 0$ if $n_0 \geq 1$.

When $(a,b) \in \{a < b+2\} \cup \{a = b+2 < -1\}$, by Theorem 4.1 (ii), if $(-1)^{n_0} \nu_{n_0}(a,b) > 0$ for $n_0 \geq 2$, then there is an $x_{n_0} \in (0,\infty)$ such that $(-1)^{n_0} \ddot{Y}_{a,b}^{(n_0)}(x) > 0$ for $x \in (0, x_{n_0})$ and $(-1)^{n_0} \ddot{Y}_{a,b}^{(n_0)}(x) < 0$ for $x \in (x_{n_0}, \infty)$; that is, $(-1)^{n_0-1} \ddot{Y}_{a,b}^{(n_0-1)}(x)$ is decreasing on $(0, x_{n_0})$ and increasing on (x_{n_0}, ∞). Thus,

$$(-1)^{n_0-1} \ddot{Y}_{a,b}^{(n_0-1)}(x) \leq \max\left\{(-1)^{n_0-1} \ddot{Y}_{a,b}^{(n_0-1)}(0), 0\right\} = \max\left\{(-1)^{n_0-1} \nu_{n_0-1}(a,b), 0\right\} = 0$$

for $x > 0$. This implies that $x \mapsto -\ddot{Y}_{a,b}(x)$ is an $n_0 - 1$-order IM function on $(0,\infty)$, namely,

$$O_{\text{incm}}\left[-\ddot{Y}_{a,b}(0,\infty)\right] = n_0 - 1.$$

Taking $n = 1$ in Theorem 4.1 and noting that $-\nu_1 = \sqrt{\pi/2}\,(2a - b)$ give the following monotonicity of $Y_{a,b}(x)$ on $(0,\infty)$.

Corollary 4.1 *(i) If* $(a,b) \in \{a \geq \max\{b+2, -1\}\}$, *then* $Y_{a,b}(x)$ *is decreasing on* $(0,\infty)$.

(ii) If $(a,b) \in \{a < b+2 \text{ and } a \leq b/2\} \cup \{a = b+2 \leq -2\}$, *then* $Y_{a,b}(x)$ *is increasing on* $(0,\infty)$.

(iii) If $(a,b) \in \{b/2 < a < b+2\} \cup \{b/2 < a = b+2 < -1\}$, *then there is an* $x_1 \in (0,\infty)$ *such that* $Y_{a,b}(x)$ *is decreasing on* $(0, x_1)$ *and increasing on* (x_1, ∞).

4.2 Special case 1: $b = a - 2$

In this case, the integral representation (3.20) becomes

$$Y_{a,a-2}(x) = a - 1 + \int_0^\infty \eta_{a,a-2}(s)\, s e^{-s^2/2} e^{-xs} ds, \tag{4.6}$$

where

$$\eta_{a,a-2}(s) = \frac{\sqrt{\pi}}{2} \sum_{n=1}^{\infty} \frac{n+a}{4^n \Gamma(n+3/2)} s^{2n}. \tag{4.7}$$

By Theorem 4.1, we derive the following corollary.

Corollary 4.2 *Let* $\ddot{Y}_{a,a-2}(x) = Y_{a,a-2}(x) - a + 1$, *where* $Y_{a,b}(x)$ *is defined in* (1.8), *and let* $\nu_n = \nu_n(a,b)$ *be defined by* (3.15). *The following statements are valid.*

(i) If and only if $a \geq -1$, *the function* $x \mapsto \ddot{Y}_{a,a-2}(x)$ *is CM on* $(0,\infty)$.

(ii) If $a < -1$, *then* $x \mapsto +(-)\ddot{Y}_{a,a-2}(x)$ *is not CM on* $(0,\infty)$. *Precisely,* $(-1)^n \ddot{Y}_{a,a-2}^{(n)}(x) \leq 0$ *for* $x \in (0,\infty)$ *and* $n \in \mathbb{N}_0$ *if and only if the inequalities* (4.2) *hold for* $b = a - 2$; *if the inequalities* (4.3) *hold for* $b = a - 2$, *then there is an* $x_n > 0$ *such that* $(-1)^n \ddot{Y}_{a,a-2}^{(n)}(x) > 0$ *for* $x \in (0, x_n)$ *and* $(-1)^n \ddot{Y}_{a,a-2}^{(n)}(x) < 0$ *for* $x \in (x_n, \infty)$.

Proof (i) If $x \mapsto \ddot{Y}_{a,a-2}(x)$ is CM on $(0,\infty)$, then

$$\lim_{x\to\infty}\left(x^4\ddot{Y}_{a,a-2}(x)\right) \geq 0.$$

By (3.8), we have $\lim_{x\to\infty}\left(x^4\ddot{Y}_{a,a-2}(x)\right) = a+1$. Hence, the necessary condition for the function $x \mapsto \ddot{Y}_{a,a-2}(x)$ to be CM on $(0,\infty)$ is: $a \geq -1$.

If $a \geq -1$, then Theorem 4.1 (i) for $b = a-2$ implies that $x \mapsto \ddot{Y}_{a,a-2}(x)$ is CM on $(0,\infty)$, which proves the sufficiency.

(ii) The second assertion follows from Theorem 4.1 (ii) for $b = a-2$.

This completes the proof. □

Remark 4.2 Let n_0 be the minimum integer such that $(-1)^n \nu_n(a,a-2) > 0$ for $n \geq n_0 \in \mathbb{N}_0$ and fixed $a \in \mathbb{R}$. Then $(-1)^{n_0}\nu_{n_0}(a,a-2) > 0$ and $(-1)^{n_0-1}\nu_{n_0-1}(a,a-2) \leq 0$ for $n_0 \geq 1$. In a similar way to Remark 4.1, if $a < -1$ and $n_0 \geq 2$, then $x \mapsto -\ddot{Y}_{a,a-2}(x)$ is an n_0-1-order IM function on $(0,\infty)$, namely,

$$O_{\text{incm}}\left[-\ddot{Y}_{a,a-2}(x)\,(0,\infty)\right] = n_0 - 1.$$

Setting $n=1$ in Corollary 4.2 and noting that $-\nu_1(a,a-2) = \sqrt{\pi/2}\,(a+2)$ give the following monotonicity of $Y_{a,a-2}(x)$ on $(0,\infty)$.

Corollary 4.3 *The following statements are valid.*

(*i*) *If* $a \geq -1$ *then the function* $Y_{a,a-2}(x)$ *is decreasing on* $(0,\infty)$.

(*ii*) *If* $a \leq -2$ *then* $Y_{a,a-2}(x)$ *is increasing on* $(0,\infty)$.

(*iii*) *If* $-2 < a < -1$, *then there is an* $x_1 \in (0,\infty)$ *such that* $Y_{a,a-2}(x)$ *is decreasing on* $(0,x_1)$ *and increasing on* (x_1,∞).

We further consider the complete monotonicity of the truncations of the asymptotic expansion (3.8):

$$D_2(x,a) = Y_{a,a-2}(x) - \left(a-1+\frac{a+1}{x^4}\right), \tag{4.8}$$

$$D_3(x,a) = Y_{a,a-2}(x) - \left(a-1+\frac{a+1}{x^4}-\frac{2(4a+3)}{x^6}\right), \tag{4.9}$$

and

$$D_4(x,a) = Y_{a,a-2}(x) - \left(a-1+\frac{a+1}{x^4}-\frac{2(4a+3)}{x^6}+\frac{3(23a+13)}{x^8}\right). \tag{4.10}$$

Theorem 4.2 *Let* $a \in \mathbb{R}$. *The following statements are valid.*

(*i*) *The function* $x \mapsto +(-)D_2(x,a)$ *is CM on* $(0,\infty)$ *if and only if* $a \leq -1$ $(a \geq -3/4)$.

(*ii*) *The function* $x \mapsto +(-)D_3(x,a)$ *is CM on* $(0,\infty)$ *if and only if* $a \geq -13/23$ $(a \leq -3/4)$.

(*iii*) *The function* $x \mapsto +(-)D_4(x,a)$ *is CM on* $(0,\infty)$ *if and only if* $a \leq -13/23$ $(a \geq -25/58)$.

Before proving Theorem 4.2, we first prove the following lemma.

Lemma 4.1 *Let*

$$P_n(a) = \frac{\sqrt{\pi}(n+a)\,n!}{2^{n+1}\Gamma(n+3/2)} - \frac{(a+1)\,n}{3} + \frac{(4a+3)\,n(n-1)}{15} - \frac{(23a+13)\,n(n-1)(n-2)}{210}. \tag{4.11}$$

Then $P_n(a) \geq (\leq)\,0$ *for all* $n \geq 4$ *if and only if* $a \leq -13/23$ ($a \geq -25/58$). *While* $-13/23 < a < -25/58$, *there is an* $n_0 > 4$ *such that* $P_n(a) > 0$ *for* $4 \leq n < n_0$ *and* $P_n(a) < 0$ *for* $n > n_0$.

Proof An arrangement yields

$$P_n(a) = -\left(\frac{n\left(23n^2-125n+172\right)}{210} - \frac{\sqrt{\pi}n!}{2^{n+1}\Gamma(n+3/2)}\right)a - \left(\frac{n\left(13n^2-81n+138\right)}{210} - \frac{\sqrt{\pi}n!}{2^{n+1}\Gamma(n+3/2)}\right) =: -c_n a - d_n,$$

where

$$c_n = \frac{n\left(23n^2-125n+172\right)}{210} - w_n,$$
$$d_n = \frac{n\left(13n^2-81n+138\right)}{210} - w_n,$$
$$w_n = \frac{\sqrt{\pi}n!}{2^{n+1}\Gamma(n+3/2)}.$$

A simple verification yields that $P_n(a) = 0$ for $n = 1, 2, 3$ and

$$P_4(a) = -\frac{4}{315}(58a+25).$$

If we prove that $c_n > 0$ for $n \geq 4$ and the sequence $\{d_n/c_n\}_{n\geq 4}$ is increasing, then

$$P_n(a) = -c_n\left(a + \frac{d_n}{c_n}\right) \geq (\leq)\,0 \text{ for all } n \geq 4$$

if and only if

$$a \leq -\lim_{n\to\infty}\frac{d_n}{c_n} = -\frac{13}{23} \text{ or } a \geq -\frac{d_4}{c_4} = -\frac{25}{58}.$$

When $-13/23 < a < -25/28$, since $\{a + d_n/c_n\}_{n\geq 4}$ is increasing with

$$\left[a + \frac{d_n}{c_n}\right]_{n=4} = a + \frac{25}{58} < 0 \text{ and } \lim_{n\to\infty}\left(a + \frac{d_n}{c_n}\right) = a + \frac{13}{23} > 0,$$

there is an $n_0 > 4$ such that $a + d_n/c_n < 0$ for $4 \leq n < n_0$ and $a + d_n/c_n > 0$ for $n > n_0$. This means that $P_n(a) > 0$ for $4 \leq n < n_0$ and $P_n(a) < 0$ for $n > n_0$.

Now, we first prove that $c_n > 0$ for $n \geq 4$. Since w_n satisfies

$$w_{n+1} = \frac{n+1}{2n+3} w_n \text{ for } n \in \mathbb{N}. \tag{4.12}$$

we have

$$\frac{c_{n+1}}{w_{n+1}} - \frac{c_n}{w_n} = \frac{1}{210} \frac{(23n+105)(n-1)(n-2)}{w_n} \geq 0 \text{ for } n \geq 2,$$

which implies the sequence $\{c_n/w_n\}_{n\geq 4}$ is increasing. It follows that

$$\frac{c_n}{w_n} \geq \frac{c_4}{w_4} = \left[\frac{1}{210} \frac{n\left(23n^2 - 125n + 172\right)}{w_n} \right]_{n=4} - 1 = 29 > 0$$

for $n \geq 4$, and hence, $c_n > 0$ for $n \geq 4$.

Second, we prove that the sequence $\{d_n/c_n\}_{n\geq 4}$ is increasing, which suffices to prove $\delta_n = c_n d_{n+1} - d_n c_{n+1} > 0$ for $n \geq 4$. By (4.12),

$$\begin{aligned} \delta_n = c_n d_{n+1} - d_n c_{n+1} &= \frac{n+1}{2n+3} w_n^2 + \frac{1}{210} \frac{(n+1)\left(23n^4 - 154n^2 + 201n - 210\right)}{2n+3} w_n \\ &\quad + \frac{1}{3150} n(n+1)\left(17n^2 - 117n + 170\right) > 0 \end{aligned}$$

for $n \geq 5$, where the inequality holds due to

$$23n^4 - 154n^2 + 201n - 210 > 0 \text{ for } n \geq 5,$$

and

$$17n^2 - 117n + 170 > 0 \text{ for } n \geq 5.$$

This together with

$$\delta_4 = \frac{5}{11} w_4^2 + \frac{287}{33} w_4 - \frac{52}{315} = \frac{12\,244}{218\,295} > 0$$

leads to $\delta_n > 0$ for $n \geq 4$. Hence, the sequence $\{d_n/c_n\}_{n\geq 4}$ is increasing.

This completes the proof. □

We are now in a position to prove Theorem 4.2.

Proof of Theorem 4.2 We only prove the assertion (iii) (others follow similarly).

Using the integral representation (4.6) and

$$\frac{1}{x^r} = \frac{1}{\Gamma(r)} \int_0^\infty s^{r-1} e^{-xs} ds \ (r > 0),$$

we rewrite (4.10) as an integral representation:

$$D_4(x,a) = \int_0^\infty \eta_{a,a-2}(s)\, s e^{-s^2/2} e^{-xs} ds - \frac{a+1}{\Gamma(4)} \int_0^\infty s^3 e^{-xs} ds$$

$$+\frac{8a+6}{\Gamma(6)}\int_0^\infty s^5 e^{-xs}ds-\frac{3(23a+13)}{\Gamma(8)}\int_0^\infty s^7 e^{-xs}ds$$
$$=\int_0^\infty I_4(s)\,se^{-s^2/2}e^{-xs}ds, \tag{4.13}$$

where

$$I_4(s)=\eta_{a,a-2}(s)-\frac{a+1}{\Gamma(4)}s^2e^{s^2/2}+\frac{8a+6}{\Gamma(6)}s^4e^{s^2/2}-\frac{3(23a+13)}{\Gamma(8)}s^6e^{s^2/2}.$$

By Bernstein Theorem, the function $x\mapsto +(-)D_4(x,a)$ is CM on $(0,\infty)$ if and only if $I_4(s)\geq(\leq)0$ for all $s>0$. If we prove that $I_4(s)\geq(\leq)0$ for all $s>0$ if and only if $a\leq -13/23$ ($a\geq -25/58$), then the required statement follows.

To this end, we expand $I_4(s)$ in power series;

$$I_4(s)=\frac{\sqrt{\pi}}{2}\sum_{n=1}^\infty\frac{n+a}{4^n\Gamma(n+3/2)}s^{2n}-\frac{a+1}{\Gamma(4)}\sum_{n=1}^\infty\frac{2n}{2^n n!}s^{2n}$$
$$+\frac{8a+6}{\Gamma(6)}\sum_{n=1}^\infty\frac{4n(n-1)}{2^n n!}s^{2n}-\frac{3(23a+13)}{\Gamma(8)}\sum_{n=1}^\infty\frac{8n(n-1)(n-2)}{2^n n!}s^{2n}$$
$$=\sum_{n=1}^\infty\frac{P_n(a)}{2^n n!}s^{2n}, \tag{4.14}$$

where $P_n(a)$ is defined by (4.11).

Now, if $a\leq -13/23$, then by Lemma 4.1, $P_n(a)>0$ for $n\geq 4$, so $I_4(s)>0$ for all $s>0$. If $a\geq -25/58$. then by Lemma 4.1, $P_n(a)<0$ for $n\geq 4$, so $I_4(s)<0$ for all $s>0$. If $-13/23<a<-25/58$, then by Lemma 4.1, $P_n(a)>0$ for $4\leq n<n_0$ and $P_n(a)<0$ for $n>n_0$, which implies that $I_4(s)=\sum_{n=1}^\infty(P_n(a)/(2^n n!))s^{2n}$ is a PN-type power series. From Lemma 2.2 it follows that there is an $s_0>0$ such that $I_4(s)>0$ for $s\in(0,s_0)$ and $I_4(s)<0$ for $s\in(s_0,\infty)$. These imply that $I_4(s)\geq(\leq)0$ for all $s>0$ if and only if $a\leq -13/23$ ($a\geq -25/58$).

This completes the proof of the third statement of this theorem. □

4.3 Special case 2: $b=2a$

In this case, $\nu_1=-\sqrt{\pi/2}(2a-b)=0$ (by Theorem 3.2), and hence, $Y'_{a,2a}(0)=Y'_{a,2a}(\infty)=0$. This implies that there is no $n\in\mathbb{N}$ with $n\geq 2$ such that $(-1)^n Y^{(n)}_{a,2a}(x)\geq(\leq)0$ for $x\in(0,\infty)$. If not, namely, there exists $n_1\in\mathbb{N}$ with $n\geq 2$ such that $(-1)^{n_1}Y^{(n_1)}_{a,2a}(x)\geq(\leq)0$ for $x\in(0,\infty)$, then $(-1)^{n_1-1}Y^{(n_1-1)}_{a,2a}(x)$ is decreasing (increasing) on $(0,\infty)$, which implies that

$$(-1)^{n_1-1}Y^{(n_1-1)}_{a,2a}(x)>(<)\lim_{x\to\infty}(-1)^{n_1}Y^{(n_1-1)}_{a,2a}(x)=0$$

for $x\in(0,\infty)$. Repeating such steps n_1-2 times gives $Y''_{a,2a}(x)>(<)0$ for $x\in(0,\infty)$, which yields

$$0=Y'_{a,2a}(0)<(>)Y'_{a,2a}(x)<(>)Y'_{a,2a}(\infty)=0$$

for all $x>0$. This leads to a contradiction.

Fortunately, since

$$\begin{aligned} Y'_{a,b}(x) &= 2x\left[\left(x^2+a+1\right)R^2(x)+(b/2-1)\,xR(x)-\frac{b}{2}\right]+(b-2a)\,R(x) \\ &= 2x\left[Y_{a+1,b/2-1}(x)-\frac{b}{2}\right]+(b-2a)\,R(x), \end{aligned}$$

we have the relation

$$\frac{1}{2x}Y'_{a,2a}(x) = Y_{a+1,a-1}(x)-a = \ddot{Y}_{a+1,a-1}(x). \tag{4.15}$$

Theorem 3.2 with this relation gives

$$\frac{1}{2x}Y'_{a,2a}(x) = \frac{1}{2}\sum_{n=0}^{\infty}(n+2)\,v_{n+2}(a.2a)\,x^n = v_0(a+1,a-1)-a+\sum_{n=1}^{\infty}v_n(a+1,a-1)\,x^n,$$

which, by comparing the coefficients of x^n, reveals

$$\begin{aligned} v_0(a+1,a-1)-a &= v_2(a.2a), \\ v_n(a+1,a-1) &= \frac{1}{2}(n+2)\,v_{n+2}(a.2a) \quad \text{for } n\geq 1 \end{aligned}$$

Now, Corollary 4.2 with the relation (4.15) gives the following corollary.

Corollary 4.4 *Let $Y_{a,b}(x)$ be defined in (1.8) and let $v_n = v_n(a,b)$ be defined by (3.15). The following statements are valid.*

(i) If and only if $a\geq -2$, the function $x\mapsto Y'_{a,2a}(x)/x$ is CM on $(0,\infty)$.

(ii) If $a<-2$, then $x\mapsto +(-)\,Y'_{a,2a}(x)/x$ is not CM on $(0,\infty)$. Precisely, $(-1)^n[Y'_{a,2a}(x)/x]^{(n)}\leq 0$ for $x\in(0,\infty)$ and $n\in\mathbb{N}$ if and only if $(-1)^n v_{n+2}(a,2a)\leq 0$; if $(-1)^n v_{n+2}(a,2a)>0$, there is an $x_n>0$ such that $(-1)^n\left[Y'_{a,2a}(x)/x\right]^{(n)}>0$ for $x\in(0,x_n)$ and $(-1)^n[Y'_{a,2a}(x)/x]^{(n)}<0$ for $x\in(x_n,\infty)$.

Remark 4.3 Let n_0 be the minimum integer such that $(-1)^n v_n(a+1,a-1)>0$ for $n\geq n_0\in\mathbb{N}_0$ and fixed $a\in\mathbb{R}$. Then $(-1)^{n_0}v_{n_0}(a+1,a-1)>0$ and $(-1)^{n_0-1}v_{n_0-1}(a+1,a-1)\leq 0$ for $n_0\geq 1$. By Remark 4.2 with the relation (4.15), if $a<-2$ and $n_0\geq 2$, then $x\mapsto -Y_{a,2a}(x)/x$ is an n_0-1-order IM function on $(0,\infty)$, namely,

$$O_{\text{incm}}\left[-\frac{Y'_{a,2a}(x)}{x}(0,\infty)\right] = n_0-1.$$

Taking $n=0$ in Theorem 4.4 and noting that

$$v_2(a,2a) = \frac{\pi}{2}+a\left(\frac{\pi}{2}+1\right)-2a = \left(\frac{\pi}{2}-1\right)\left(a+\frac{\pi}{\pi-2}\right)$$

give the following monotonicity of $Y_{a,2a}(x)$ on $(0,\infty)$.

Corollary 4.5 *Let*

$$a_1 = -\frac{\pi}{\pi-2} = -2.751.... \tag{4.16}$$

The following statements are valid.

(*i*) *If* $a \geq -2$, *then the function* $Y_{a,2a}(x)$ *is increasing on* $(0,\infty)$.

(*ii*) *If* $a \leq a_1$, *then* $Y_{a,2a}(x)$ *is decreasing on* $(0,\infty)$.

(*iii*) *If* $a_1 < a < -2$, *then there is an* $x_0 > 0$ *such that* $Y_{a,2a}(x)$ *is increasing on* $(0,x_0)$ *and decreasing on* (x_0,∞).

5 Functional inequalities

In this section, we will establish some new functional inequalities for R using those higher order monotonicity presented in Sect. 3.

Recall that the integral representation of remainder term of Taylor formula:

$$f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k = \int_{x_0}^{x} \frac{(x-t)^{n-1}}{(n-1)!} f^{(n)}(t)\,dt = \frac{(x-x_0)^n}{(n-1)!}\int_0^1 (1-u)^{n-1} f^{(n)}(\xi)\,du, \tag{5.1}$$

where $\xi = ux + (1-u)x_0$. Applying (5.1) to the function $Y_{a,b}(x)$ with $x_0 = 0$ yields

$$Y_{a,b}(x) - \sum_{k=0}^{n-1} \nu_k(a,b)\,x^k = \frac{x^n}{(n-1)!}\int_0^1 (1-u)^{n-1} Y_{a,b}^{(n)}(\xi)\,du$$

for $n \in \mathbb{N}$. By Theorem 4.1 and $Y_{a,b}^{(n)}(x) = \ddot{Y}_{a,b}^{(n)}(x)$ for $n \geq 1$, we can obtain the following inequalities

Proposition 5.1 *Let* $m_1, m_2 \in \mathbb{N}_0$, $\nu_0 = a\pi/2$ *and* $\nu_k \equiv \nu_k(a,b)$ *for* $k \geq 1$ (*defined by* (3.15)). *The following statements are valid*:

(*i*) *If* $(a,b) \in \{a \geq \max\{b+2,-1\}\}$, *then*

$$\sum_{k=0}^{2m_1+1} \nu_k(a,b)\,x^k < Y_{a,b}(x) < \sum_{k=0}^{2m_2} \nu_k(a,b)\,x^k \quad (x>0).$$

(*ii*) *If* $(a,b) \in \{a < b+2\} \cup \{a = b+2 < -1\}$ *and* $(-1)^n \nu_n(a,b) \leq 0$ *for* $n \in \mathbb{N}$, *then*

$$(-1)^n Y_{a,b}(x) < (-1)^n \sum_{k=0}^{n-1} \nu_k(a,b)\,x^k \quad (x>0).$$

(*iii*) *If* $(a,b) \in \{a < b+2\} \cup \{a = b+2 < -1\}$ *and* $-\nu_1(a,b) > 0$, *then*

$$Y_{a,b}(x) < \max\left\{\frac{a\pi}{2}, b+1\right\} \quad (x>0).$$

Setting $b = a-2$ in Proposition 5.1 gives the following corollary.

Corollary 5.1 *Let* $m_1, m_2 \in \mathbb{N}_0$, $\nu_0 = a\pi/2$ *and* $\nu_k \equiv \nu_k(a,b)$ *for* $k \geq 1$ (*defined by* (3.15)). *The following statements are valid*:

(*i*) *If* $a \geq -1$ *then*

$$\sum_{k=0}^{2m_1+1} \nu_k(a,a-2)\,x^k < Y_{a,a-2}(x) < \sum_{k=0}^{2m_2} \nu_k(a,a-2)\,x^k \quad (x>0). \tag{5.2}$$

(*ii*) *If* $a < -1$ *and* $(-1)^n v_n(a, a-2) \le 0$ *for* $n \in \mathbb{N}$, *then*

$$(-1)^n Y_{a,a-2}(x) < (-1)^n \sum_{k=0}^{n-1} v_k(a, a-2) x^k \ (x > 0).$$

(*iii*) *If* $a < -1$ *and* $-v_1(a, a-2) > 0$, *i.e.*, $-2 < a < -1$, *then*

$$Y_{a,a-2}(x) < \max\left\{\frac{a\pi}{2}, a-1\right\} \ (x > 0).$$

Using the relation (4.15), the inequality

$$Y_{a+1,a-1}(x) < (>) \sum_{k=0}^{n-1} v_k(a+1, a-1) x^k$$

is equivalent to

$$\frac{1}{2x} Y'_{a,2a}(x) < (>) \sum_{k=0}^{n-1} v_k(a+1, a-1) x^k - a,$$

which leads to

$$Y_{a,2a}(x) - Y_{a,2a}(0) = \int_0^x Y'_{a,2a}(t)\, dt < (>) \sum_{k=0}^{n-1} 2v_k(a+1, a-1) \int_0^x t^{k+1} dt - 2a \int_0^x t dt.$$

Then by Corollary 5.1, we have the following inequalities.

Corollary 5.2 *Let* $m_1, m_2 \in \mathbb{N}_0$, $v_0 = (a+1)\pi/2$ *and* $v_k \equiv v_k(a, b)$ *for* $k \ge 1$ *be defined by* (3.15). *The following statements are valid*:

(*i*) *If* $a \ge -2$, *then for* $x > 0$,

$$\frac{a\pi}{2} - ax^2 + \sum_{k=0}^{2m_1+1} \frac{2v_k(a+1, a-1)}{k+2} x^{k+2} < Y_{a,2a}(x) < \frac{a\pi}{2} - ax^2 + \sum_{k=0}^{2m_2} \frac{2v_k(a+1, a-1)}{k+2} x^{k+2}. \quad (5.3)$$

(*ii*) *If* $a < -2$, $(-1)^n v_n(a+1, a-1) \le 0$ *for* $n \in \mathbb{N}$, *then for* $x > 0$,

$$(-1)^n Y_{a,2a}(x) < (-1)^n \frac{a\pi}{2} - (-1)^n ax^2 + (-1)^n \sum_{k=0}^{n-1} \frac{2v_k(a+1, a-1)}{k+2} x^{k+2}.$$

(*iii*) *If* $a < -2$ *and* $-v_1(a+1, a-1) > 0$, *namely*, $-3 < a < -2$, *then for* $x > 0$,

$$Y_{a,2a}(x) < \frac{a\pi}{2} + \frac{x^2}{2} \max\{a(\pi - 2) + \pi, 0\}$$

By Theorem 4.2, $D_3(x, a) > (<) 0$ and $D_4(x, a) < (>) 0$ for $x > 0$ if $a \ge \max\{-25/58, -13/23\}$ ($a \le \min\{-3/4, -13/23\}$), leading to the following statement.

Proposition 5.2 *If $a \geq -25/58$, then for $x > 0$,*

$$a-1+\frac{a+1}{x^4}-\frac{2(4a+3)}{x^6} < Y_{a,a-2}(x) < a-1+\frac{a+1}{x^4}-\frac{2(4a+3)}{x^6}+\frac{3(23a+13)}{x^8}.$$

The double inequality reverses if $a \leq -3/4$.

Proposition 5.2 with the relation (4.15) and

$$\int_x^\infty Y'_{a,2a}(t)\,dt = 2a+1-Y_{a,2a}(x)$$

give the following inequalities.

Proposition 5.3 *If $a \geq -83/58$, then for $x > 0$,*

$$2a+1-\frac{a+2}{x^2}+\frac{4a+7}{x^4}-\frac{23a+36}{x^6} < Y_{a,2a}(x) < 2a+1-\frac{a+2}{x^2}+\frac{4a+7}{x^4}.$$

The double inequality reverses if $a \leq -7/4$.

6 Known and new bounds for $R(x)$

Solving those functional inequalities given in Corollaries 5.1, 5.2, Propositions 5.2 and 5.3 yields some bounds for $R(x)$. The expressions of these bounds are often somewhat complicated, and the ranges might just be one or more sub-interval of $(0,\infty)$ because of the non-negativity of the radicand. We list two examples here.

Example 1 Taking $a = -2$ and $m_2 = 1$ in (5.3) yields

$$\left(x^2-2\right)R^2-4xR < x^4-\frac{1}{3}\sqrt{2\pi}x^3+\left(2-\frac{1}{2}\pi\right)x^2-\pi$$

for $x > 0$. The inequality is equivalent to

$$\begin{aligned}&\left(x^2-2\right)R^2-4xR-x^4+\frac{1}{3}\sqrt{2\pi}x^3-\frac{4-\pi}{2}x^2+\pi\\ &=\left(\left(x^2-2\right)R-\frac{4x+\sqrt{4x^6-4\sqrt{2\pi}x^5/3-2\pi x^4+8\sqrt{2\pi}x^3/3+8\pi}}{2}\right)\\ &\quad\times\left(R-\frac{6\pi-3(4-\pi)x^2+2\sqrt{2\pi}x^3-6x^4}{12x+\sqrt{36x^6-12\sqrt{2\pi}x^5-18\pi x^4+24\sqrt{2\pi}x^3+72\pi}}\right)<0.\end{aligned}$$

The first factor can be written as

$$x(xR-1)-2R-x-\frac{\sqrt{4x^6-4\sqrt{2\pi}x^5/3-2\pi x^4+8\sqrt{2\pi}x^3/3+8\pi}}{2},$$

which is negative for $x > 0$ due to $R' = xR-1 < 0$. Then

$$R(x) > \frac{6\pi-3(4-\pi)x^2+2\sqrt{2\pi}x^3-6x^4}{12x+\sqrt{36x^6-12\sqrt{2\pi}x^5-18\pi x^4+24\sqrt{2\pi}x^3+72\pi}} = B_1(x) \tag{6.1}$$

for $0 < x \leq 1.492...$, where 1.492... is the zero of the numerator of the above fraction.

Example 2 For another example, taking $a = -1$ in Proposition 5.3 yields

$$-\frac{2\left(x^8 - x^2 + 15\right)}{x^8} < Y_{-1,-3}(x) = \left(x^2 - 1\right)R^2 - 3xR < -\frac{2\left(x^6 - 1\right)}{x^6}$$

for $x > 0$. Solving the above inequalities gives

$$R(x) > \frac{4}{x^3}\frac{x^6 - 1}{3x^4 + \sqrt{x^8 + 8x^6 + 8x^2 - 8}} = B_2(x) \tag{6.2}$$

for $x \geq 1$, and

$$R(x) < \frac{4}{x^4}\frac{x^8 - x^2 + 15}{3x^5 + \sqrt{x^{10} + 8x^8 + 8x^4 - 128x^2 + 120}} = B_3(x) \tag{6.3}$$

for $x \geq 1.273...$ or $0 < x \leq 1.067...$, where $1.273...$ and $1.067...$ are zeros of the polynomial $x^{10} + 8x^8 + 8x^4 - 128x^2 + 120$.

If $Y_{a,b}(x)$ has constant bounds, then we can obtain simple bounds for $R(x)$ on $(0, \infty)$. To this end, we define

$$\dot{Y}_{a,b}(x) = Y_{a,b}(x) - Y_{a,b}(0) = \left(x^2 + a\right)R^2 + bxR - \frac{a\pi}{2}, \tag{6.4}$$

and recall (4.1), i.e.,

$$\ddot{Y}_{a,b}(x) = Y_{a,b}(x) - Y_{a,b}(\infty) = \left(x^2 + a\right)R^2 + bxR - (b + 1).$$

Now, using Corollaries 4.3 and 4.5 then solving the inequality $\dot{Y}_{a,b}(x) < (>)\, 0$ or $\dot{Y}_{a,b}(x) < (>)\, 0$, we establish several known and new bounds for $R(x)$.

Corollary 6.1 *The following inequalities hold for* $x > 0$:

$$\frac{\pi}{\sqrt{(9 - 2\pi)x^2 + 2\pi} + 3x} < R(x) < \frac{4}{\sqrt{x^2 + 8} + 3x}, \tag{6.5}$$

$$\frac{3}{\sqrt{x^2 + 6} + 2x} < R(x) < \frac{\pi}{\sqrt{(4 - \pi)x^2 + 2\pi} + 2x}, \tag{6.6}$$

$$\frac{\pi}{\sqrt{x^2 + 2\pi} + (\pi - 1)x} < R(x) \tag{6.7}$$

Remark 6.1 The upper bound in (6.5) is $U_S(x)$ defined in (1.7), which was established by Sampford [6]; the lower bound in (6.7) is $L_B(x)$ defined in (1.6), which was proved by Boyd [13]; the lower bound given in (6.6) was due to [8]. The lower bound in (6.5) and upper bound in (6.6) seem to be new.

Proof of Corollary 6.1 (i) Taking $a = -1$ in Corollary 4.3 (i) reveals that the function $x \mapsto Y_{-1,-3}(x)$ is decreasing on $(0, \infty)$, which yields that

$$-2 < Y_{-1,-3}(x) = \left(x^2 - 1\right)R^2 - 3xR < -\frac{\pi}{2}$$

for $x > 0$; that is, $\dot{Y}_{-1,-3}(x) < 0$ and $\ddot{Y}_{-1,-3}(x) > 0$. Factoring $\dot{Y}_{-1,-3}(x)$ yields

$$\dot{Y}_{-1,-3}(x) = \left(\left(x^2-1\right)R - \frac{\sqrt{(9-2\pi)x^2+2\pi}+3x}{2}\right) \times \left(R - \frac{\pi}{\sqrt{(9-2\pi)x^2+2\pi}+3x}\right).$$

The first factor can be written as

$$x(xR-1) - R - \frac{x+\sqrt{2(4-\pi)x^4+8\pi}}{2},$$

which is negative for $x > 0$ due to $R' = xR - 1 < 0$. Then the inequality $\dot{Y}_{-1,-3}(x) < 0$ implies the first one of (6.5). Similarly, the inequality $\ddot{Y}_{-1,-3}(x) > 0$ implies the second one of (6.5).

(ii) Taking $a = -2$ in Corollary 4.3 (ii) reveals that the function $Y_{-2,-4}(x)$ is increasing on $(0,\infty)$, which yields that

$$-\pi < Y_{-2,-4}(x) = \left(x^2-2\right)R^2 - 4xR < -3 \tag{6.8}$$

for $x > 0$; that is, $\dot{Y}_{-2,-4}(x) > 0$ and $\ddot{Y}_{-2,-4}(x) < 0$. Factoring $\dot{Y}_{-2,-4}(x)$ leads to

$$\dot{Y}_{-2,-4}(x) = \left(\left(x^2-2\right)R - 2x - \sqrt{(4-\pi)x^2+2\pi}\right)\left(R - \frac{\pi}{\sqrt{(4-\pi)x^2+2\pi}+2x}\right).$$

The first factor can be written as

$$x(xR-1) - 2R - x - \sqrt{(4-\pi)x^2+2\pi},$$

which is negative for $x > 0$ due to $R' = xR - 1 < 0$. Then the inequality $\dot{Y}_{-2,-4}(x) > 0$ implies the second one of (6.6). Similarly, the inequality $\ddot{Y}_{-2,-4}(x) < 0$ implies the first one of (6.6).

(iii) Taking $a = a_3 = -2/(\pi-2) = -1.751...$ in Corollary 4.3 (iii) yields

$$Y_{a_3,a_3-2}(x) = \left(x^2+a_3\right)R^2 + (a_3-2)xR < \max\left\{\frac{a_3\pi}{2}, a_3-1\right\} = -\frac{\pi}{\pi-2},$$

which is equivalent to

$$\left((\pi-2)x^2-2\right)R^2 - 2(\pi-1)xR + \pi < 0$$

for $x > 0$. Factoring yields

$$\left(\frac{\pi}{R} - (\pi-1)x + \sqrt{x^2+2\pi}\right)\left(\frac{1}{R} - \frac{(\pi-1)x+\sqrt{x^2+2\pi}}{\pi}\right) < 0$$

for $x > 0$. Since the first factor can be written as

$$\pi\left(\frac{1}{R} - x\right) + x + \sqrt{x^2+2\pi},$$

which is positive for $x > 0$ due to $R < 1/x$, the inequality means that

$$\frac{1}{R} - \frac{(\pi - 1)x + \sqrt{x^2 + 2\pi}}{\pi} < 0$$

for $x > 0$; that is, the inequality (6.7) holds.

This completes the proof. □

Corollary 6.2 *The following inequalities*

$$\frac{\pi}{\sqrt{2(4-\pi)x^2 + 2\pi} + 2x} < R(x) < \frac{\pi + 2}{\sqrt{4x^2 + \pi(\pi + 2)} + \pi x}, \tag{6.9}$$

$$R(x) < \frac{\pi}{\sqrt{(\pi - 2)^2 x^2 + 2\pi} + 2x} \tag{6.10}$$

hold for $x > 0$.

Remark 6.2 The lower bound in (6.9) is $L_{GU}(x)$ defined in (1.6), which was first presented by Gasull and Utzet [8]; the upper bound in (6.10) is $U_B(x)$ defined in (1.7), which was proved by Boyd [13]. The upper bound in (6.9) seems to be new.

Proof of Corollary 6.2 (i) Taking $a = a_1 = -\pi/(\pi - 2)$ in Corollary 4.5 (ii) reveals that the function $Y_{a_1,2a_1}(x)$ is decreasing on $(0, \infty)$, which yields that

$$2a_1 + 1 < Y_{a_1,2a_1}(x) = \left(x^2 + a_1\right)R^2 + 2a_1 xR < \frac{a_1\pi}{2}$$

for $x > 0$; that is, $\dot{Y}_{a_1,2a_1}(x) < 0$ and $\ddot{Y}_{a_1,2a_1}(x) > 0$. Factoring $\ddot{Y}_{a_1,2a_1}(x)/R^2$ yields

$$\frac{\ddot{Y}_{a_1,2a_1}(x)}{R^2} = \left(\frac{\pi + 2}{(\pi - 2)R} - \frac{\pi x}{\pi - 2} + \frac{\sqrt{4x^2 + \pi(\pi + 2)}}{\pi - 2}\right) \times \left(\frac{1}{R} - \frac{\sqrt{4x^2 + \pi(\pi + 2)} + \pi x}{\pi + 2}\right),$$

where the first factor is positive since it can be written as

$$\frac{\pi}{\pi - 2}\left(\frac{1}{R} - x\right) + \frac{2}{(\pi - 2)R} + \frac{\sqrt{4x^2 + \pi(\pi + 2)}}{\pi - 2} > 0$$

for $x > 0$. Hence, the inequality $\ddot{Y}_{a_1,2a_1}(x) > 0$ implies the second one of (6.9). Similarly, factoring $\dot{Y}_{a_1,2a_1}(x)/R^2$ yields

$$\frac{\dot{Y}_{a_1,2a_1}(x)}{R^2} = \left(\frac{\pi^2}{2(\pi - 2)}\frac{1}{R} - \frac{2\pi x - \pi\sqrt{2(4-\pi)x^2 + 2\pi}}{2(\pi - 2)}\right) \times \left(\frac{1}{R} - \frac{2x + \sqrt{2(4-\pi)x^2 + 2\pi}}{\pi}\right),$$

where the first factor is positive since it is equivalent to

$$\frac{\pi^2}{2(\pi-2)}\left(\frac{1}{R}-x\right)+\frac{1}{2}\pi x+\frac{\pi\sqrt{2(4-\pi)x^2+2\pi}}{2(\pi-2)}>0$$

for $x>0$. Then, the inequality $\dot{Y}_{a_1,2a_1}(x)<0$ implies the first one of (6.9).

(ii) Taking $a=a_2=-2/(4-\pi)=-2.329...$ in Corollary 4.5 (iii) yields

$$Y_{a_2,2a_2}=\left(x^2+a_2\right)R^2+2a_2xR>\frac{a_2\pi}{2}=2a_2+1=-\frac{\pi}{4-\pi},$$

which is equivalent to $\ddot{Y}_{a_2,2a_2}(x)>0$. Factoring $\ddot{Y}_{a_2,2a_2}(x)$ yields

$$\frac{\ddot{Y}_{a_2,2a_2}(x)}{R^2}=\left(\frac{\pi}{4-\pi}\frac{1}{R}-\frac{2x-\sqrt{(\pi-2)^2x^2+2\pi}}{4-\pi}\right)\times\left(\frac{1}{R}-\frac{2x+\sqrt{(\pi-2)^2x^2+2\pi}}{\pi}\right),$$

where the first factor is equivalent to

$$\frac{\pi}{4-\pi}\left(\frac{1}{R}-x\right)+\frac{(\pi-2)x+\sqrt{(\pi-2)^2x^2+2\pi}}{4-\pi}>0$$

for $x>0$. Then the inequality $\ddot{Y}_{a_2,2a_2}(x)>0$ implies (6.10)

This completes the proof. □

Finally, as noted in Introduction, the bounds for the Mills ratio $R(x)$ derived from the higher order monotonicity of $Y_{a,b}(x)$ are in general more accurate than those derived from the monotonicity of $Y_{a,b}(x)$. Now we illustrate this claim in the form of remarks.

Remark 6.3 From the proofs of Corollaries 6.1 and 6.2, the lower bound in (6.9), i.e., $L_{GU}(x)$, has the highest accuracy for small x among all bounds given in Corollaries 6.1 and 6.2. Precisely,

$$R(x)-L_{GU}(x)=R(x)-\frac{\pi}{\sqrt{2(4-\pi)x^2+2\pi}+2x}=\frac{2(\pi-3)}{3\pi}x^3+O\left(x^4\right).$$

On the other hand, by (6.1),

$$R(x)-B_1(x)=\frac{11}{60}x^5+O\left(x^6\right).$$

Hence, as approximations of $R(x)$, $B_1(x)$ has higher accuracy than $L_{GU}(x)$ for small x.

Remark 6.4 Similarly, the upper bound in (6.5), i.e., $U_S(x)$, has the highest accuracy for large x among all bounds given in Corollaries 6.1 and 6.2, which satisfies

$$R(x)-U_S(x)=R(x)-\frac{4}{\sqrt{x^2+8}+3x}=-\frac{2}{x^7}+O\left(\frac{1}{x^9}\right).$$

On the other hand, the bounds $B_2(x)$ and $B_3(x)$ given in (6.2) and (6.3) satisfy

$$R(x) - B_2(x) = \frac{30}{x^9} + O\left(\frac{1}{x^{11}}\right),$$
$$R(x) - B_3(x) = -\frac{396}{x^{11}} + O\left(\frac{1}{x^{13}}\right).$$

Therefore, as approximations of $R(x)$, $B_2(x)$ is weaker than $B_3(x)$ and better than $U_S(x)$ for large x. More precisely, we can show that

$$R(x) < \frac{4}{x^4}\frac{x^8 - x^2 + 15}{3x^5 + \sqrt{x^{10} + 8x^8 + 8x^4 - 128x^2 + 120}} < \frac{4}{\sqrt{x^2+8} + 3x}$$

for $x \geq 3.9$, and

$$R(x) - \frac{4}{x^3}\frac{x^6 - 1}{3x^4 + \sqrt{x^8 + 8x^6 + 8x^2 - 8}} < \frac{4}{\sqrt{x^2+8} + 3x} - R(x)$$

for $x \geq 1.05$.

7 Concluding remarks

This paper derived the asymptotic expansion, Maclaurin series, and integral representation of $Y_{a,b}(x) = \left(x^2 + a\right)R^2(x) + bxR(x)$, and established its higher-order monotonicity using sign rules for NP-type power series and Laplace integrals. As applications, we obtained several functional inequalities and new bounds for the Mills ratio $R(x)$.

Next we present two remarks.

Remark 7.1 By Theorem 4.2, the function $x \mapsto \ddot{Y}_{a,a-2}(x) = Y_{a,a-2}(x) - a + 1$ is CM on $(0,\infty)$ if and only if $a \geq -1$. As a result, the function $\ddot{Y}_{-1,-3}(x) = \left(x^2 - 1\right)R^2(x) - 3xR(x) + 2$ is CM on $(0,\infty)$. Interestingly, this function is related to Sampford's [6] results that $0 < (1/R(x))' < 1$ and $(1/R(x))'' > 0$ for $x \in \mathbb{R}$. In fact, since

$$\left(\frac{1}{R(x)}\right)'' = \frac{\left(x^2-1\right)R^2(x) - 3xR(x) + 2}{R^3(x)} = \frac{Y_{-1,-3}(x) + 2}{R^3(x)},$$
$$(\ln R(x))''' = \left(x - \frac{1}{R(x)}\right)'' = -\left(\frac{1}{R(x)}\right)'',$$

we see that the function $R^3(1/R)''$ and $-R^3(\ln R)'''$ are CM on $(0,\infty)$.

Remark 7.2 The convexity of $x \mapsto 1/R(x)$ on $(0,\infty)$ yields

$$\frac{2}{\pi} < \left(\frac{1}{R}\right)' = \frac{1 - xR}{R^2} < 1$$

for $x > 0$. The left hand side inequality gives an upper bound for R:

$$R(x) < \frac{2}{\sqrt{x^2 + 8/\pi} + x} \text{ for } x > 0.$$

This bound was due to Pollak [29].

Moreover, inspired by Theorem 4.2, we propose the following conjecture.

Conjecture 7.1 *Let*

$$D_n(x,a) = Y_{a,a-2}(x) - \left(a - 1 + \sum_{k=1}^{n} \frac{(-1)^k u_k}{x^{2k}}\right),$$

where u_k is defined by (3.7). *Then $x \mapsto (-1)^n D_n(x,-1)$ is CM on $(0,\infty)$.*

Finally, we emphasize that the two sign rules for the NP-type power series and Laplace integrals, that is, Lemmas 2.2 and 2.3, play important roles, where the former has been applied in the study for certain special functions, see for example, [30, Lemma 7], [31–35]. We believe that the latter, Lemma 2.3, will also be widely applied in special functions.

Acknowledgements
Not applicable.

Author contributions
All the authors contributed equally to this paper.

Funding information
No funding was received to assist with the preparation of this manuscript.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Use of AI tools
The authors have not used Artificial Intelligence (AI) tools in the creation of this paper.

Competing interests
The authors declare no competing interests.

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Received: 8 December 2025 Accepted: 8 June 2026 Published online: 19 June 2026

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